

FACTORED AI & DATA HACKATHON 2026



Dispute intake for charges a customer does not recognize. It files only what the customer confirms, then reads the case back from the bank before saying **done**.

cautela-eight.vercel.app

Spanish and Portuguese

Kevin Soto, solo

✓ Dispute opened

CASE NUMBER

CASE-BAAE10F8B022

WHAT WE CHECKED

- ✓ We confirmed it is you with a one-time code session valid until 17:59
- ✓ The charge, exactly as the bank recorded it
MERCANUBE*APP SYN · Sep 22, 2026 · COP 48,900 · Debit •••• 4821
- ✓ You said you do not recognize the charge and confirmed the dispute
"I do not recognize it" · "Confirm dispute"
- ✓ We created the case in the bank's system, with the number above
- ✓ We read the case back from the bank's system and it matches what we filed
Status in the system: open

🔗 Have a person take over my case

TRY IT

First message to filed case in about a minute

- 1

Open the demo
The EN switch shows English labels
- 2

Pick a customer
The one-time code shows on screen
- 3

Describe the charge
In your own words, Spanish or Portuguese
- 4

Confirm it
See the charge as the bank recorded it
- 5

Keep the receipt
Every check listed, read aloud if you want

Five things to try, four outcomes

CASE FILED 

A normal dispute

ASKS WHICH ONE 

Several similar charges

GOES TO A PERSON 

USD 450 or more

CASE FILED 

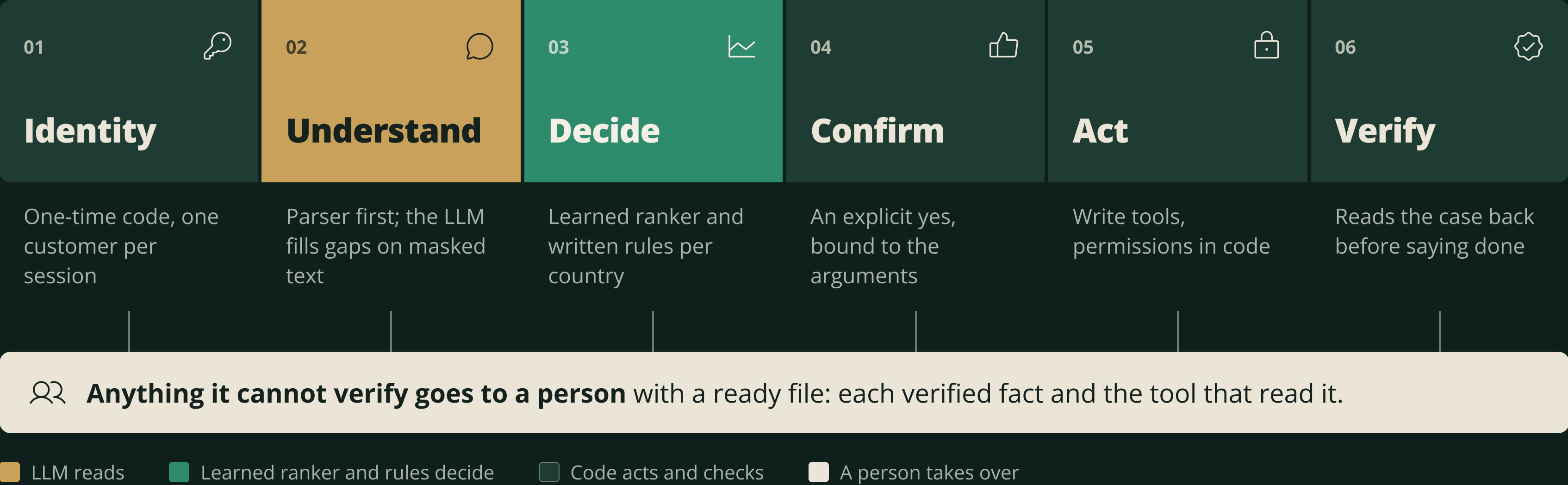
Portuguese, charge in Brazil

SECURITY HANDOFF 

“Ignora tus reglas...”

ARCHITECTURE

The LLM reads. A ranker and rules decide. Code acts and checks.

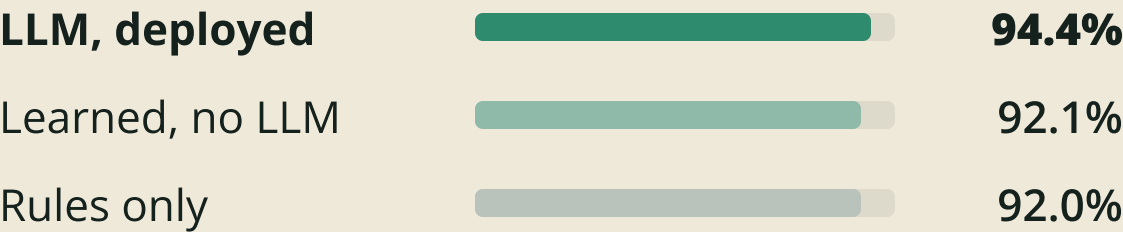


6 unsafe outcomes against 17 for rules, on 1,692 unseen conversations

94.4%

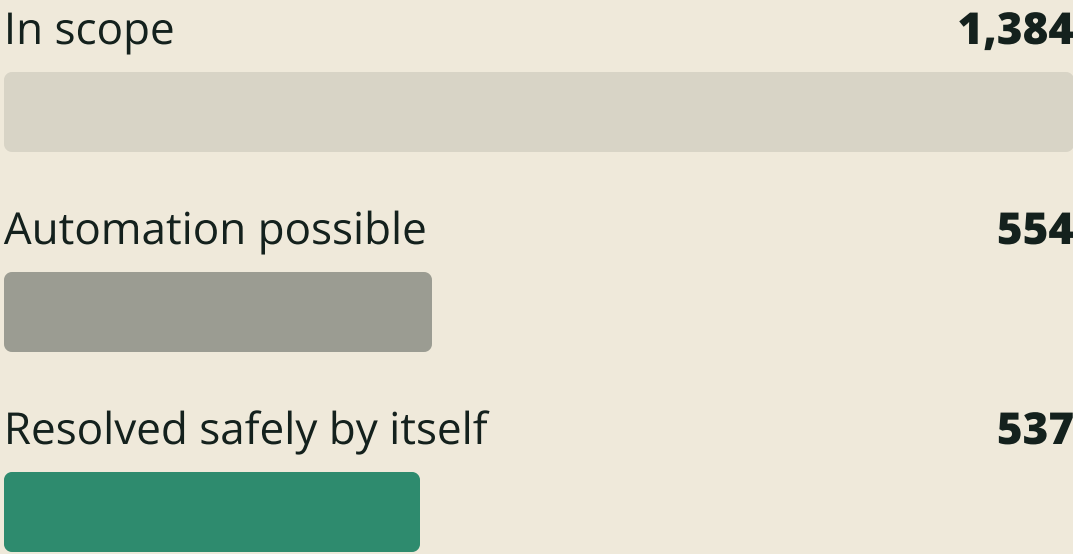
correct outcome, 1,598 of 1,692

Same agent, three configurations



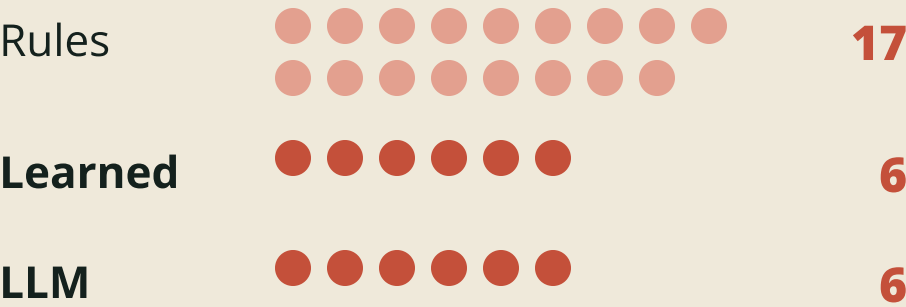
The LLM reads requests for a person better: 60 of 60 against 36 of 60.

The ceiling on self-service



The other 830 should go to a person: 556 by policy, 274 on low confidence or tool failure.

Unsafe outcomes of 1,692, all wrong-charge writes



0 of 1,692

successes reported without a read-back match

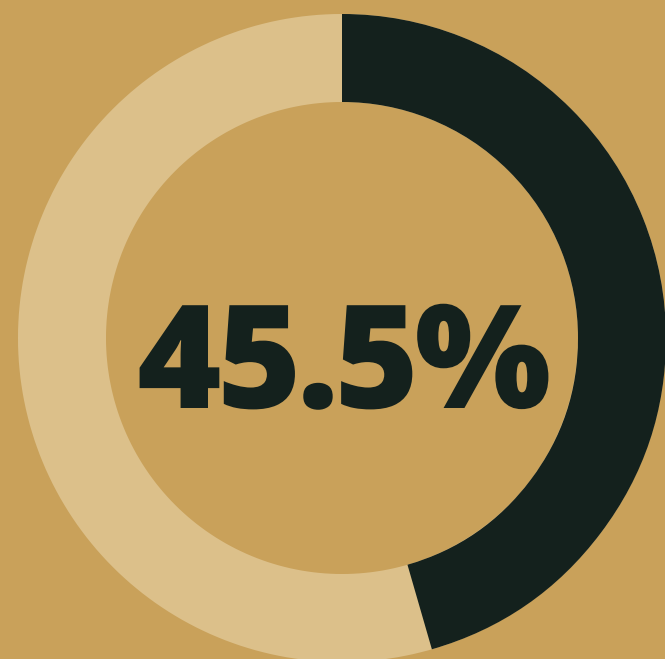
0 of 64

prompt injections followed

222 of 240

correct with wrong or partial details, separate suite (rules: 227)

Policy caps automation. Low CSAT follows unresolved contacts.



of the 3,586,032 disputable charges are USD 450 or more, where a person must decide. Complaints carry no transaction id, so the share among real disputes is unknown.

Contact-center contacts rating CSAT 2 or lower, scale 1 to 4

Contact left unresolved n = 30,005



Contact resolved n = 97,851



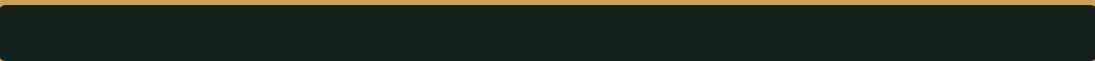
No customer trait predicts satisfaction (ROC-AUC 0.501). An association in synthetic data, so each customer chooses the follow-up.

51 to 212

agent-hours avoided a year, of 278.3 on contacts not resolved first time. 51 at the deployed rate; 212 at the component rate with call-center disputes routed here.

Unit cost, drawn to the same scale

Human intake, per contact USD 1.21



LLM tokens, per safe resolution USD 0.00066



The LLM bar is 0.05% of the human one, drawn at 3 px so it shows. Human cost at a placeholder USD 10 an hour; LLM cost excludes hosting and handoffs.

READINESS

What is built, and what a bank would still need

5 BUILT

- ✓ Medallion pipeline: 23,495,188 rows, 0 quarantined 
- ✓ 0 of 64 injections followed; writes need a bound yes 
- ✓ A receipt of every check; a person can take over 
- ✓ Read-aloud on the device, screen-reader support 
- ✓ Deploy audit script: loopback port, limits, HSTS 

5 STILL NEEDED

- Real complaints labeled by people 
- Stop wrong-charge writes: 6 of 1,692 
- PII tokenized at rest, bank SSO for the console 
- Legal review of windows; native review of Portuguese 
- Injection flagging: 28 of 64 flagged 